



Experiment 2

Prepared By:

Name: Mahin Raj

Course: BCA (2nd Semester)

SAP ID: 590025881

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Submitted To:

Mr. Vibhu Gautam

Faculty, School of Computer Science

UPES, Dehradun



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mahin-mr12 commit

9eed2b6 · 3 weeks ago



421 lines (421 loc) · 8.53 KB

Preview

Code

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1. Declare these variables (x, y, and z) as integers. Assign a value of 9 to x. Assign a value of 7 to y, perform addition, multiplication, division and subtraction on these two variable and print out the results

```
In [1]: x = 8
        y = 7
        print("Addition:",x+y)
        print("Multiplication", x*y)
        print("Division", x/y)
        print("Subtration", x-y)
```

```
Addition: 15
Multiplication 56
Division 1.1428571428571428
Subtration 1
```

2. Write a program where the radius is taken as input to compute the area of a circle

```
In [ ]: r = int(input("Enter the radius of a circle"))
        area = 3.14 * r * r
        print("Area of the circle:",area)
```

```
Area of the circle: 28.259999999999998
```

3. Write a python program to solve $(x+y)*(x+y)$

```
In [ ]: x = int(input("Enter the value of x"))
        y = int(input("Enter the value of y"))
        z = (x+y)*(x+y)
        print(z)
```

```
49
```

6 Write a program to compute the length of the hypotenuse (c) of a right

6. Write a program to compute the length of the hypotenuse (c) of a right triangle using Pythagoras theorem

```
In [ ]: import math
h = int(input("Enter the height of the triangle"))
b = int(input("Enter the base of the triangle"))
sum = (h * h) + (b * b)
c = math.sqrt(sum)
print(c)
```

5.0

7. Write a program to find simple interest

```
In [ ]: p = int(input("Enter the principle"))
t = int(input("Enter the time of the loan"))
r = int(input("Enter the rate of interest"))
si = (p * r * t)/100
print(si)
```

3000.0

8. Write a program to find the area of triangle when lengths of sides are given

```
In [ ]: import math
a = float(input("Enter side a: "))
b = float(input("Enter side b: "))
c = float(input("Enter side c: "))
s = (a + b + c) / 2
area = math.sqrt(s * (s - a) * (s - b) * (s - c))
print(area)
```

0.0

9. Write a program to convert given seconds into hours, minutes and remaining seconds.

```
In [ ]: time = int(input("Enter the seconds"))
        hour = time // 3600
        rem_time = time % 3600
        min = rem_time // 60
        sec = rem_time % 60
        print(hour, "hour", min, "minutes", sec, "seconds")
```

1 hours 34 minutes 38 seconds

10. Write a program to swap two numbers without taking additional variable.

```
In [5]: a = int(input("Enter the first number"))
        b = int(input("Enter the second number"))
        a = a + b
        b = a - b
        a = a - b
        print(a,b)
```

9 5

11. Write a program to find the sum of first n natural numbers.

```
In [8]: n = int(input("Enter the value to find the sum of numbers till that number"))
        sum = n * (n+1) // 2
        print(sum)
```

15

12. Write a program to print truth table for bitwise operators (&, |, and ^ operators)

```
In [9]: print("A B  A&B  A|B  A^B")
        print("-----")
        for a in range(2):
            for b in range(2):
                and op = a & b
```

```

or_op = a | b
xor_op = a ^ b

print(a, b, " ", and_op, " ", or_op, " ", xor_op)

```

A	B	A&B	A B	A^B
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

13. Write a program to find the left shift and right shift values of a given number

```

In [4]: num = int(input("Enter a number"))
        shift = int(input("Enter the number of shifts"))
        left_shift = num << shift
        right_shift = num >> shift
        print(left_shift)
        print(right_shift)

```

32
0

```

In [13]: num = [10,20,56,78,89]
        print(10 in num)
        print(30 in num)
        print(40 not in num)

```

True
False
True

15. Using membership operator find whether a given character is in a string